

Reviewer Pack — Minimal Rank of Tropicalized Sheaves over Manifolds vs AC0 P...

Entry 94f57b0d8684 · INCONCLUSIVE

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Minimal Rank of Tropicalized Sheaves over Manifolds vs AC0 Parity Circuit Size

Entry ID: 94f57b0d8684 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 04:40:48 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Tropical Geometry) × Topology (Manifold Theory) **Field B** (complexity object): Complexity Theory: AC⁰ Circuit Complexity

Statement:

[‘For any n , there exists a polynomial-time computable invariant $\psi(T)$ on the set of tropicalized sheaves T over an n -dimensional manifold M such that $\psi(T) \geq c \cdot \log(\text{size}(T))$ for any tropicalized sheaf T computing PARITY on n inputs, and ψ is invariant under the action of automorphisms of M .’, ‘The polynomial-time computable invariant ψ satisfies $\psi(T) > 2c \cdot \log(n)$ for any AC⁰ parity circuit C with size less than $2^{(1.5n)}$.’, ‘If there exists an AC⁰ parity circuit C with size less than $2^{(1.5n)}$, then the rank of the corresponding tropicalized sheaf T over M is greater than $c' \cdot \log(\text{size}(C))$.’]

Rationale (proposer’s reasoning):

[‘Tropical geometry provides a way to study algebraic structures in a combinatorial fashion, which can potentially reveal hidden patterns related to computational complexity.’, ‘Manifolds provide a rich structure to study the behavior of tropicalized sheaves, which might expose new insights into circuit complexity, particularly for AC⁰ parity circuits.’, ‘By establishing a quantitative relation between these two fields, we could potentially find new invariants that can be used to classify or lower bound the complexity of AC⁰ parity circuits.’]

Taxonomy category: AC0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2df1a6eb464523c3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The polynomial-time computable invariant $\psi(T)$ on tropicalized sheaves and $\psi(C)$ on AC⁰ parity circuits is compared against predefined thresholds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.70	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropicalized sheaves" AND "AC0 circuit complexity" - "minimal rank" AND "tropical geometry" AND AC0 - "PARITY circuit" AND Manifold Theory AND complexity theory

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

# Function to generate a random n-dimensional manifold
def generate_manifold(n):
    return [random.randint(1, 10) for _ in range(n)]

# Function to construct a tropicalized sheaf over a manifold
def construct_tropicalized_sheaf(manifold):
    return sum(manifold)

# Function to compute the size of a tropicalized sheaf
def size_of_sheaf(sheaf):
    return len(sheaf)

# Function to compute the rank of a tropicalized sheaf (simplified for testing)
def rank_of_sheaf(sheaf):
    return max(sheaf) - min(sheaf) + 1

# Polynomial-time computable invariant  $\psi(T)$  on tropicalized sheaves
def invariant_psi(sheaf):
    return size_of_sheaf(sheaf)

# Function to generate a random AC0 parity circuit with size less than  $2^{(1.5n)}$ 
def generate_ac0_circuit(n):
    return [random.choice([0, 1]) for _ in range(2 ** (int(1.5 * n)))]

# Function to compute the size of an AC0 parity circuit
def size_of_circuit(circuit):
```

```

    return len(circuit)

# Main function to run a single trial with a given seed
def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Parameters for testing
    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        manifold = generate_manifold(n)
        sheaf = construct_tropicalized_sheaf(manifold)
        psi_value = invariant_psi(sheaf)

        # Check the conjecture for tropicalized sheaves computing PARITY
        if psi_value < 2 * math.log(n):
            results.append({
                "metric_name": "Invariant  $\psi(T)$ ",
                "metric_value": psi_value,
                "instances_tested": 1,
                "conjecture_holds": False,
                "counterexample": f" $\psi(T)={psi\_value} \leq 2c \cdot \log(n)={2 * math.log(n)}$ "
            })
        else:
            results.append({
                "metric_name": "Invariant  $\psi(T)$ ",
                "metric_value": psi_value,
                "instances_tested": 1,
                "conjecture_holds": True,
                "counterexample": ""
            })

    # Generate and check AC0 parity circuits
    for _ in range(3): # Test with multiple instances of AC0 circuits
        circuit = generate_ac0_circuit(n)
        psi_value_circuit = invariant_psi(circuit)

        if psi_value_circuit > 2 * math.log(n):
            results.append({
                "metric_name": "Invariant  $\psi(C)$ ",
                "metric_value": psi_value_circuit,
                "instances_tested": 1,
                "conjecture_holds": True,
                "counterexample": ""
            })
        else:
            results.append({
                "metric_name": "Invariant  $\psi(C)$ ",
                "metric_value": psi_value_circuit,
                "instances_tested": 1,
                "conjecture_holds": False,
                "counterexample": f" $\psi(C)={psi\_value\_circuit} \leq 2c \cdot \log(n)={2 * math.log(n)}$ "
            })

    # Compute mean and std of metric values

```

```

psi_values = [r["metric_value"] for r in results if "Invariant  $\psi(T)$ " in r]
invariant_psi_mean = sum(psi_values) / len(psi_values)
invariant_psi_std = (sum((x - invariant_psi_mean) ** 2 for x in psi_values) / len(psi_values))

# Check if the conjecture holds for all seeds
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

return {
    "seed": seed,
    "invariant_psi_mean": invariant_psi_mean,
    "invariant_psi_std": invariant_psi_std,
    "support_fraction": support_fraction
}

if __name__ == "__main__":
    import sys

    if not sys.argv[1:]:
        seeds = [2*i + 7 for i in range(5, 30)] # List of 30 primes
    else:
        seeds = list(map(int, sys.argv[1:]))

    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    # Compute overall metrics and support fraction
    invariant_psi_means = [r["invariant_psi_mean"] for r in results]
    invariant_psi_stds = [r["invariant_psi_std"] for r in results]
    support_fractions = [r["support_fraction"] for r in results]

    overall_invariant_psi_mean = sum(invariant_psi_means) / len(invariant_psi_means)
    overall_invariant_psi_std = (sum((x - overall_invariant_psi_mean) ** 2 for x in invariant_psi_means) / len(invariant_psi_means)) ** 0.5
    overall_support_fraction = sum(support_fractions) / len(support_fractions)

    if all(r["support_fraction"] == 1 for r in results):
        print(f"RESULT: SUPPORTED mean={overall_invariant_psi_mean} std={overall_invariant_psi_std} support={overall_support_fraction}")
    elif overall_support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={overall_invariant_psi_mean} std={overall_invariant_psi_std} support={overall_support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["support_fraction"] == 1), None)
        print(f"RESULT: FALSIFIED counterexample=\"support_fraction<{overall_support_fraction}\"")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac675d60.py", line 125, in <module>

```

trial_result = run_trial(seed)
^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac675d60.py", line 57, in run_trial
    psi_value = invariant_psi(sheaf)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac675d60.py", line 36, in invariant_psi
    return size_of_sheaf(sheaf)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ac675d60.py", line 28, in size_of_sheaf
    return len(sheaf)
    ^^^^^^^^^
TypeError: object of type 'int' has no len()

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture's conditions. | next: Investigate and fix the error in the test code to allow for a proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 12

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12274
2	propose	ollama_remote	glm4:latest	0	0	11945
3	propose	ollama_remote	glm4:latest	0	0	12524
4	propose	ollama_remote	glm4:latest	0	0	12058
5	preregistration	ollama_remote	glm4:latest	0	0	5278
6	novelty	ollama_remote	glm4:latest	0	0	4741
7	novelty	ollama_remote	glm4:latest	0	0	5441
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14129
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8990
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9095
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15981
12	judge	ollama_remote	glm4:latest	0	0	8449

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 120906 ms total latency. Provider mix: {'ollama_remote': 12}

(full prompt+response transcripts available in `research/audit/94f57b0d8684.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/94f57b0d8684.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/94f57b0d8684.tar.gz` (if generated)